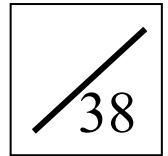


Name: _____

Date: _____



1. Find the sum of all the negative integers x satisfying the inequality $x > -2\pi$.

A. -21π
B. -21
C. -5
D. 0

☐

2. If $2x \leq 3$, which of the following is not true?

A. $4x \leq 2x + 3$
B. $1 - 2x \leq -2$
C. $2x + 3 \leq 6$
D. $\frac{2x}{3} \leq 1$

☐

3. If $x \leq y$ and $z > y$, where z is a negative number, which of the following is not true?

A. $x \leq y < z$
B. $y - x < z - x$
C. $xz < xy$
D. $-\frac{x}{4} < -\frac{z}{4}$

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4. Solve the inequality $3(x + 2) < 5(6 - x)$.

A. $x > 3$
B. $x < 3$
C. $x > -3$
D. $x < -3$

☐

5. Which of the following groups of numbers can all satisfy the inequality $7x + 3 > 17$?

- A. $-1, 0, 1$
- B. $1, 3, 5$
- C. $2, 3, 4$
- D. $2.1, 3.2, 4.3$



6. (a) Solve the inequality $\frac{x}{4} - 2 \leq 0.8 - \frac{x}{3}$.

(3 marks)

(b) Write down all positive integers satisfying the inequality in (a).

(2 marks)

7. Write down all positive integers satisfying the inequality $\frac{4x+3}{5} < \frac{1-3x}{6} + 4$.

(5 marks)

- *8. A **non-negative** number x satisfies the equation $\frac{2x-k}{3} = \frac{5x-7}{4} + \frac{x+k}{6}$, where k is a positive integral constant. How many possible values of k are there? (8 marks)